Project details

Application team

SAFETECH GLOBAL LIMITED

Organisation details

Type	Business

Team members

Full name	Email	EDI survey
Fatma Erturk	hello@otshield.io	Complete
Fatma Erturk	fatma.erturk@otshield.io	Complete

Application details

Competition name
Contracts for Innovation: Cyber scale in critical sectors

Application name
OTShield

When do you wish to start your project?
1 September 2026

Project duration in months
12 months

Has this application been previously submitted to Innovate UK?
No

Who made you aware of the competition?
Other

How long has the company been established for?
Established 1 to 5 years

What is your organisation's primary area of focus?
Energy

Project and scope summary

Project and scope summary

OTShield is an integrated operational technology (OT) and industrial control systems (ICS) security platform for the energy sector, built around two cooperating modes that share one data layer. SOC Mode provides deception based active defence and threat hunting. Research Mode provides an offline analyst copilot. This project deploys and operates both modes inside a real UK energy operator's live OT environment over 12 months, taking the platform from TRL 7 to TRL 8.

The problem. Energy Critical National Infrastructure (substations, grid control, generation, refining) depends on OT and ICS mixing modern equipment with legacy controllers that cannot be patched or taken offline. Conventional IT security is blind to OT protocols, and the asset visibility tools operators already deploy (Dragos, Claroty) show what is on the network but not who is attacking it or how. The result is a detection gap exactly where a sophisticated actor inside an energy network does the most damage, at a time when operators must evidence intrusion detection and response under the NIS Regulations, NIS2 equivalent expectations and the NCSC CAF.

What makes it innovative. OTShield combines two capabilities that asset visibility tools lack, delivered to coexist with them rather than replace them. SOC Mode deploys realistic, protocol aware decoy HMIs and controllers (Siemens WinCC, Rockwell, Schneider and generic profiles) speaking Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3 and IEC 60870-5-104, with Purdue aware deep packet inspection, auto discovered asset inventory and full MITRE ATT&CK for ICS coverage. Because legitimate plant traffic never touches a decoy, any interaction is a high confidence, low noise signal of reconnaissance or intrusion, and the platform detects and hunts with no risk to live controllers. Research Mode adds an offline copilot for analysts: a research workbench (Library, Summary, Inventory, Ports and Services, Threads, Findings, Vulns) with a local LLM that summarises evidence, answers questions with cited sources, and turns attacker activity into auditable Findings. The two modes share one data layer, so live attacker behaviour from SOC Mode informs research, and research insight informs defence. Real attacker tactics from OTShield's live internet exposed decoy fabric feed the in plant platform via signed update bundles, so detection reflects current energy ICS threats.

How it tackles the different aspects of the challenge, as an integrated solution. A single deployment addresses all four scope challenges. First, proactive cyber defence of CNI systems including incident response, delivered by SOC Mode through high confidence detection and SIEM and SOAR integrated alerting. Second, detection and eviction of sophisticated actors from CNI networks, by SOC Mode surfacing protocol aware adversary behaviour (unauthorised writes, out of range values, interlock override attempts, remote system discovery) and mapping it across the ATT&CK for ICS kill chain. Third, threat hunting for CNI organisations, made practical by Research Mode, where analysts query detections, manuals and asset data together and follow leads safely against the operator's real asset base. Fourth, surfacing legacy system vulnerabilities in CNI, where SOC Mode inventories and risk scores legacy controllers and Research Mode reasons over them to show how an attacker would reach and abuse them. These are one platform on a common data layer, so detection informs hunting and hunting informs defence.

Scope, readiness and what the project delivers. Both modes are operational today. SOC Mode and the decoy fabric run on an internet exposed deployment validated against real ICS attackers, and Research Mode runs as a working offline copilot, placing the capability at TRL 7. Within the project we deploy both modes into one energy operator's live OT environment, tune them to the operator's assets, protocols and legacy equipment, run continuous detection, threat hunting and incident response with their SOC and analyst teams, capture metrics on detection rate, time to detect and analyst time saved, evidence operator acceptability, complete a commercialisation business model and plan, and analyse the standards and homologation (NIS, NIS2, IEC 62443, NCSC CAF) needed to sell into the sector. By month 12 the innovation will have demonstrated user acceptability and market fit in a real operational environment, been improved on the basis of end user feedback, and produced a completed business plan and development case study. The work is original, demonstrated in a genuine operational environment, and a deployable product, not a framework, methodology or consultancy. Well over 50% of contract value is attributed directly and exclusively to R&D.

Lead organisation. Safetech Global Limited (trading as OTShield, Companies House 15233187, 7 Bell Yard, London WC2A 2JR, England & Wales) is the sole UK lead and contracting legal entity. The application has been discussed and agreed within the organisation.

Public description

Public description

This project strengthens UK energy cyber security by demonstrating a new way to detect and respond to attackers inside the operational technology (OT) and industrial control systems (ICS) that run grid control, substations, generation and refining.

Energy infrastructure depends on control systems that mix modern equipment with older legacy controllers that cannot easily be patched or taken offline.

Conventional IT security tools are not designed for these environments, and the monitoring tools many operators use show what is on a network but give limited insight into who is attacking it and how. This leaves a gap in detecting a skilled intruder quickly and responding before harm is done.

OTShield is an integrated OT security platform with two complementary modes. The first, an active defence mode, places realistic decoy versions of industrial devices across the network. Because normal plant activity never interacts with these decoys, any contact is a strong, low noise signal of reconnaissance or intrusion, giving security teams clear, high confidence alerts with no risk to the live equipment that keeps the lights on. The second, an offline analyst assistant mode, helps the operator's analysts make sense of what the platform sees: it summarises technical evidence, answers questions with cited sources, and helps them hunt for threats and find where legacy systems may be exposed. The two modes share one foundation, so active defence informs the analyst's investigation and the analyst's findings sharpen future detection.

Over twelve months, OTShield will deploy and operate both modes inside a real energy operator's environment, working closely with their security team. The work will tune the technology to the operator's real systems and protocols, run detection and threat hunting exercises, and measure how well it performs in day to day operation, including how much analyst time it saves. The project will also produce a plan for bringing the technology to market and an analysis of the relevant security standards for the sector.

By the end the aim is a proven, operator accepted capability demonstrated in a genuine operational setting, ready for wider adoption across the energy sector and adjacent infrastructure such as water and transport. The outcome is stronger national resilience, better protection for essential public services, and a competitive UK product.

OTShield is delivered by Safetech Global Limited, a UK company specialising in OT and ICS security, with experienced UK based specialists. All project work is carried out in the UK.

Applicant location

Applicant location

Lead organisation

Safetech Global Limited (trading as OTShield)

Company registration number: 15233187 (Companies House, England & Wales)

Registered address: 7 Bell Yard, London, WC2A 2JR, United Kingdom

Safetech Global Limited is the sole lead and contracting legal entity for this project. All key project work and deliverables are carried out in the United Kingdom.

Application questions

1. Critical sectors (not scored)

Critical sectors (not scored)

Energy

2. Challenges (not scored)

Challenges (not scored)

This project addresses all four challenges. A single OTShield deployment in an energy operator's OT environment delivers proactive CNI defence and incident response, detection and eviction of sophisticated actors from CNI networks, threat hunting for CNI organisations, and surfacing of legacy system vulnerabilities, all from one deception based active defence platform sharing a common data layer.

3. Animal testing (not scored)

Will your project involve any trials with animals or animal testing?

No

4. Permits and licences (not scored)

Will you have the correct permits and licences in place to carry out your project?

Not applicable

5. International collaboration (not scored)

Does your proposed work involve any international collaboration or engagement?

Answer yet to be provided

6. Export licence (not scored)

Export licence (not scored)

No

7. Trusted Research and Innovation (not scored)

Trusted Research and Innovation (not scored)

This project relates to UKRI's Trusted Research and Innovation Principles, and we have considered each area below.

Dual use applications. OTShield is a dual use technology. Its primary application is defensive: protecting civilian critical national infrastructure (energy, water, transport) from cyber attack. The same deception based detection and threat hunting capability also has non civilian relevance to national security and defence customers undertaking OT and ICS protection. The project develops protective and detective capability only. It does not develop offensive cyber tooling, exploits or attack capability.

Relevance to NSI Act areas. The work is relevant to several of the 17 sensitive areas of the UK National Security and Investment Act, principally Energy (the target sector), Critical Suppliers to Government, Communications, Transport, Data Infrastructure and Artificial Intelligence.

Export control. No export control licence is required to carry out this project. All work, participants and deliverables are located in the United Kingdom, and there is no transfer of controlled technology, software or data outside the UK during the project. No export control applications are therefore in place or pending. Any future international commercialisation would be assessed separately against the relevant export control guidance before any export.

Strategic Export Control List. The project does not involve any physical items or substances on the UK Strategic Export Control List. The software developed will be reviewed against the relevant cyber security and information security controls before any future export.

Safeguards. Safetech Global Limited follows NPSA and NCSC Trusted Research and Secure Innovation guidance, retains all project data and IP in the UK, and is willing to undertake BPSS or equivalent personnel checks. The company and its UK based subcontractors comply with UK trade sanctions and arms embargoes, and the project has no involvement with sanctioned countries or entities.

8. Proposed idea or technology

How does the project meet the challenge described in the competition scope?

Proposed technology. OTShield is an integrated OT and ICS security platform for the energy sector, built around two cooperating modes that share one data layer. SOC Mode provides deception based active defence and threat hunting: realistic, protocol aware decoy HMIs and controllers (Siemens WinCC, Rockwell, Schneider, generic) speaking Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3 and IEC 60870-5-104, with Purdue aware deep packet inspection, asset inventory and full MITRE ATT&CK for ICS coverage. Research Mode provides an offline copilot for analysts: a research workbench (Library, Summary, Inventory, Ports and Services, Threads, Findings, Vulns) with a local LLM that summarises evidence, answers questions with citations and turns attacker activity into findings. Live attacker behaviour from SOC Mode informs research, and research insight informs defence.

How it meets the challenge. A single deployment in an energy operator's real environment meets all four scope challenges. SOC Mode delivers proactive CNI defence and incident response through high confidence, low noise decoy detections, and detects and helps evict sophisticated actors by surfacing protocol aware intent (unauthorised writes, out of range values, interlock overrides, remote discovery) mapped across the ICS kill chain. Research Mode makes threat hunting practical for CNI organisations: analysts query detections, manuals and asset data together, follow leads safely, and surface how legacy systems could be reached and abused. The platform coexists with, rather than replaces, the asset visibility tools (Dragos, Claroty) operators already run, adding both the active deception layer and the analyst copilot those tools lack.

Current state of development and readiness. The platform is operational today. SOC Mode and the decoy fabric run on an internet exposed deployment validated against real ICS attackers, and Research Mode runs as a working offline copilot, placing the capability at TRL 7. This project takes it to TRL 8 by deploying both modes inside a UK energy operator's live OT environment, tuning them to the operator's assets, protocols and legacy equipment, running detection and threat hunting campaigns with their SOC and analyst teams, and evidencing acceptability. Real attacker tactics from the live decoy fabric feed the in plant platform via signed update bundles, so detection and hunting reflect current energy ICS threats. The result is a proven, operator accepted capability that meets the challenge in a genuine operational setting.

[OTShield_Q8_Appendix_v3.pdf (opens in a new window) (/application/10206957/form/question/54227/forminput/158004/file/956871/download)](/application/10206957/form/question/54227/forminput/158004/file/956871/download).

9. Technical project summary

What are the main technical challenges you are addressing?

How we address the challenge. OTShield is an integrated energy OT/ICS security platform with two cooperating modes sharing one data layer. SOC Mode provides deception based active defence: protocol-aware decoy HMIs and controllers (Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3, IEC 60870-5-104), Purdue-aware deep-packet inspection, asset inventory and MITRE ATT&CK for ICS. Research Mode provides an offline analyst copilot (Library, Summary, Inventory, Threads, Findings, Vulns) with a local LLM. We deploy both modes into a live energy operator's environment and prove them against the operator's real assets and threats.

What the innovation is. Two things current OT tools lack, delivered together and coexisting with Dragos/Claroty. First, decoy based detection where any decoy interaction is hostile by construction, giving high confidence, low noise signals with zero risk to live controllers. Second, an offline copilot that turns detections, manuals and asset data into cited, auditable findings and makes safe threat hunting practical. A shared data layer means live attacker behaviour informs research and research informs defence.

Main technical deliverables. A deployed, operator-accepted platform (both modes) running in the live OT environment; energy-specific decoys and protocol dissectors; SIEM/SOAR integration; tuned detection and threat-hunting analytics; a TRL 8 evidence pack; and full documentation including a signed software bill of materials.

R&D proving scientific, environmental and commercial merit. The R&D is solution exploration and design for the energy deployment, development of energy-protocol decoys and analytics, and prototyping and field-testing in a representative operational environment, evaluated against adjudicated criteria: detection rate of adjudicated intrusions at least 95%, false positives at most 1%, mean time to detect at most five minutes, and analyst time saved. Scientific merit is shown by measured detection and calibration; environmental merit by protecting the energy systems society depends on and avoiding the disruption of a successful attack; commercial merit by an operator-accepted product with a completed commercialisation plan. Over 50% of value is R&D.

What deploying it achieves. By month 12 the operator gains a proven capability that detects sophisticated actors in its OT network, supports incident response and safe threat hunting, and surfaces legacy exposure, all without risk to live plant. The result is stronger UK energy resilience and a deployable product ready to scale across the sector and adjacent critical infrastructure.

10. Current state of the art and intellectual property

Are similar products currently available in the market?

Current state of the art. The OT security market is led by asset-visibility and detection platforms (Dragos, Claroty, Nozomi). They are strong at telling operators what is on the network and raising anomaly alerts, but they depend on a human analyst or a SOAR playbook to act, and they do not deploy active deception. Separately, IT-centric deception vendors (CounterCraft, Acalvio, Fidelis) offer decoys, but rarely with native energy-protocol fidelity, an integrated offline analyst copilot, or a live ICS attacker feed.

How we differentiate. OTShield is a single integrated platform combining two capabilities the incumbents do not offer together. SOC Mode delivers deception based active defence with protocol-aware decoys (Modbus, S7Comm, EtherNet/IP, OPC UA, DNP3, IEC 60870-5-104) where any decoy interaction is hostile by construction, giving high confidence, low noise detection with zero risk to live controllers. Research Mode adds an offline analyst copilot that turns detections, manuals and asset data into cited, auditable findings and makes safe threat hunting practical. The two modes share one data layer; live attacker tactics from our internet-exposed decoy fabric keep both current. We coexist with, not replace, Dragos/Claroty.

Novel concepts, approaches and tools. Novel concepts: hostile-by-construction decoy detection in energy OT, and a copilot that fuses real network behaviour with documentation. Novel approaches: a shared data layer so defence informs research and research informs defence; signed offline update bundles that keep an air-gapped plant current. Tools and technologies: protocol-aware dissectors and decoys, MITRE ATT&CK for ICS mapping, and an offline local LLM, all built on open-source components (Apache 2.0 / MIT) to avoid licence lock-in.

Existing IP and freedom to operate. Existing OTShield IP (the decoy fabric, dissectors, asset platform and research workbench) is owned by Safetech Global and brought to the project at no charge. All third-party components are permissively licensed open-source, so freedom to operate is clear and there is no closed-source dependency.

Handling IP during the project and subcontractors. New IP generated in the project is owned by Safetech Global, recorded in an IP register, and the Authority is granted the licence required by the contract. A signed software bill of materials tracks all components and licences. Subcontractors (specialist skills only) work under written terms that assign project IP to Safetech Global and flow down the contract's IP and confidentiality requirements, maintaining freedom to operate.

11. Project plan and methodology

Describe your project plan and identify the main milestones.

Resources. Delivery is led by Fatma Erturk with a Senior Cyber and AI Verification Engineer, a new UK based OT Detection and Deception Engineer for in-plant deployment, and a Cyber Security Advisor for independent assurance. Non-labour covers field hardware, a Cisco-compatible appliance, an evaluation workstation, an independent security review and travel to the energy operator. Existing OTShield IP (both modes) is brought at no charge.

Methodology and management. We run four quarterly phases with monthly delivery syncs, a shared continuously maintained risk register, and a formal go/no-go gate at the end of each quarter against the success criteria. If a minimum metric is missed by more than 20%, scope is renegotiated jointly. Fatma is accountable and is the single point of contact with Innovate UK and the operator. Work starts only after contract approval.

Establishing technical and commercial feasibility. Technical feasibility is established by deploying both modes in the live operator environment and measuring against adjudicated criteria. Commercial feasibility is established through operator acceptance, a completed commercialisation business model and plan, and a standards and homologation analysis (NIS, NIS2, IEC 62443, NCSC CAF).

Main success criteria (SMART). Detection of at least 95% of adjudicated OT-protocol intrusions; false-positive rate at most 1% on legitimate traffic; mean time to detect at most five minutes; net positive analyst time saved in at least two of three review sessions; and formal operator acceptance, all evidenced by week 52.

Milestones and payments (4 x GBP 75,000, matching the finance summary). M1, end of month 3: both modes deployed and tuned in the operator environment, signed test plan (payment GBP 75,000). M2, end of month 6: detection and threat-hunting analytics field-tested, interim metrics (GBP 75,000). M3, end of month 9: continuous operation, success metrics captured, independent security review (GBP 75,000). M4, end of month 12: TRL 8 qualification, operator acceptance, commercialisation plan and case study, handover (GBP 75,000).

Risks and mitigation. Technical: detection efficacy against advanced actors, mitigated by live attacker TTPs and adversarial validation. Commercial: slow operator procurement, mitigated by engagement letters and a wider pipeline. Operational/environmental: any impact to live controllers, mitigated because decoys are isolated, traffic never touches them, and all changes run under the operator's change control plus independent security review.

[OTShield_Q11_ProjectPlan.pdf (opens in a new window) (/application/10206957/form/question/54230/forminput/158022/file/957050/download)](/application/10206957/form/question/54230/forminput/158022/file/957050/download).

12. Technical team and expertise

Who is in the technical team? What expertise do they offer?

Fatma Erturk, Technical Lead and Project Delivery. Founder and CEO of OTShield and lead architect of both SOC Mode and Research Mode. 14+ years of software engineering and 8+ years of hands-on OT/ICS protocol expertise (Modbus, IEC 104, DNP3, S7). Former SCADA software engineer building real industrial process-control systems, and SCADA Blue Team member at NATO Locked Shields 2019. Recognised as Best SCADA/ICS/OT Security Training Provider 2024. Fatma designed the deception fabric, the protocol-aware extraction pipeline and the offline research workbench, so she directly delivers the energy-specific decoys, dissectors and analytics at the core of this project.

Jon Medvenics, Senior Cyber and AI Verification Engineer. UK Chartered Cyber Security Professional (ChCSP), OSCP, GXPN, GREM. Government delivery for the Ministry of Defence (classified-network SOC), Ministry of Justice (Red Team Operator), NHS Digital (incident response) and Parliamentary Digital Service. He leads detection-engineering and the adversarial validation that proves detection efficacy against advanced ICS-aware actors, and underpins the safe threat-hunting capability in Research Mode.

Alan Jenkins, Cyber Security Advisor and Assurance. 35+ years in cyber and enterprise security; 20+ years as a senior RAF officer (Squadron Leader); graduate of the UK Defence Intelligence and Security School. Former CISO at Babcock International, T-Systems UK and CSC UK, and CISO-in-Residence at CyLon. He provides independent assurance over the architecture, threat model, data handling and the operator engagement, which is critical for deploying safely inside live energy CNI.

OT Detection and Deception Engineer. Recruited for in-plant deployment, energy-protocol decoy tuning and SOC/SIEM integration, ensuring the team has dedicated capacity for the operational work in the operator's environment.

Why this team delivers the benefits. Between them the team has built and operated the platform, holds chartered and offensive-security credentials, has delivered into UK government and defence environments, and has hands-on NATO-grade ICS defence experience. All key work is carried out in the UK by named individuals with directly relevant, demonstrated experience in exactly the protocols, environments and assurance standards this project requires.

13. Costs and value for money

How much will the project cost?

Answer yet to be provided

14. Commercial potential

What is the commercial potential of your project?

Customer need. Energy operators must detect and respond to intruders inside OT networks under the NIS Regulations, NIS2 equivalent expectations and the NCSC CAF, but their current tools alert without acting and their analysts are scarce and overloaded. They need high confidence detection that is safe in a live plant, plus a way to make their analysts more effective. OTShield meets both needs in one platform: SOC Mode for deception based active defence, Research Mode for an offline analyst copilot.

Commercial potential and timescales. OTShield is a commercial dual-use product, not a bespoke build. It is pre-revenue but commercially defined, with three published price tiers and an active pipeline of UK and EU critical-infrastructure operators. This project delivers the operator-accepted, TRL 8 evidence and the commercialisation plan during the 12 months, with paid commercial deployments expected to ramp from project end as the energy proof point converts the pipeline.

Delivery plan and route to market. We sell directly and through channel partners, positioning OTShield as a high-margin OT security offering, deployable as a Cisco-compatible appliance or software-only to fit the customer's stack. The energy operator in this project becomes the reference deployment and case study that anchors sales into the rest of the sector. Adoption is fast (deploy in weeks, coexisting with existing tooling) which shortens the sales cycle.

Competitive advantage over the nearest solutions. The nearest available solutions are asset-visibility and detection platforms (Dragos, Claroty, Nozomi). They tell operators what is happening but depend on a human or a SOAR playbook to act, and they do not deploy active deception. IT-centric deception vendors exist but rarely with native energy-protocol fidelity, an integrated offline analyst copilot, or a live ICS attacker feed. OTShield combines deception based active defence and an offline copilot in one platform, coexisting with the incumbents rather than replacing them. That combination, with near-zero false positives and no risk to live controllers, is the significant advantage.

Existing relationships and future potential. We are selected for CyLon, Digital Catapult Cyber 101 and Cyber Runway in the UK, the Portuguese Air Force InnCyber programme and Germany's Hubgrade Cyber. Beyond energy, the same capability addresses water and transport CNI, defence supply-chain and ICS-vendor red teams, across UK public and private sectors and, in time, Five Eyes and NATO-aligned international markets. The energy proof point is directly transferable to each.

[Return to your project finances (/application/10206957/form/section/21721/)](/application/10206957/form/section/21721/) to complete or make changes to your organisation's financial information.

Project cost breakdown

SAFETECH GLOBAL LIMITED £0

Organisation

[View finances (/application/10206957/form/FINANCE)](/application/10206957/form/FINANCE)

Labour	£0
Overheads	£0

Materials	£0
Capital usage	£0
Subcontracting	£0
Travel and subsistence	£0
Other costs	£0
Total VAT	£0

Totals **£0**

Labour	£0
Overheads	£0
Materials	£0
Capital usage	£0
Subcontracting	£0
Travel and subsistence	£0
Other costs	£0
Total VAT	£0

Payment milestones

Month	Milestone	% of project costs	Payment request
Total payment requested		0.00%	£0

Supporting information

Project impact

Understanding the benefits of the projects Innovate UK supports

These organisations have not completed the project impact survey:

- SAFETECH GLOBAL LIMITED

This application cannot be submitted until all partners complete the survey.

Partner	Status
SAFETECH GLOBAL LIMITED (Lead)	Incomplete

Terms and conditions

Award terms and conditions

The following organisations have not yet accepted:

- SAFETECH GLOBAL LIMITED

This application cannot be submitted until all partners accept our terms and conditions.

Partner	Terms and conditions	Status
SAFETECH GLOBAL LIMITED (Lead)	Procurement (/application/10206957/form/terms-and-conditions/organisation/137834/question/54095)	Not yet accepted